

MODEL COMPRESSION

Problem Statement

Modern machine learning models often deliver high accuracy but at the cost of large size, slow inference, and heavy resource usage. This makes them difficult to deploy on edge devices, mobile phones, and low-latency systems, which have limited memory, processing power, and strict real-time requirements. As a result, highly accurate models in lab may still be impractical in real-world applications. Model compression techniques help reduce those effects.

Description of methods used

Prunning

Pruning works by identifying and removing weights that have minimal influence on the model’s predictions. These weights are typically very small in magnitude or rarely activated during inference, meaning they contribute little to the final output. By setting such low-impact weights to zero, pruning effectively removes unnecessary connections while preserving the model’s overall performance.

Quantization

Quantization reduces numerical precision in neural networks, which typically use 32-bit floating-point (float32) numbers. It converts weights to lower-precision formats, such as float16 or int8(used in this project), by approximating exact values. This reduces both memory usage and computational cost while maintaining similar model performance.

Results and interpretation

Model	Size (MB)	Inference Time (ms/sample)	Accuracy (%)
Original	12.46	1.21	98.74
Pruned	12.46	1.14	98.63
Quantized	3.17	1.09	98.74

Model Size: Pruning didn’t reduce size, but quantization cut it by almost a quarter.

Inference Time: Both improved; pruning ~7% faster, quantization ~10% faster.

Accuracy: Pruning caused a slight drop, trading accuracy for speed, while quantization reduced size and inference time without losing accuracy.

Overall, quantization proved to be the more effective method, reducing model size and inference time while maintaining accuracy.